Class 05: Data Viz with ggplot

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Background

Today we are going to learn all about the **ggplot2** package in R.

There are tones of graphics packages in R. These include "base R graphics" (like the `plot()` function we used last day) and add on packages like **ggplot2**.

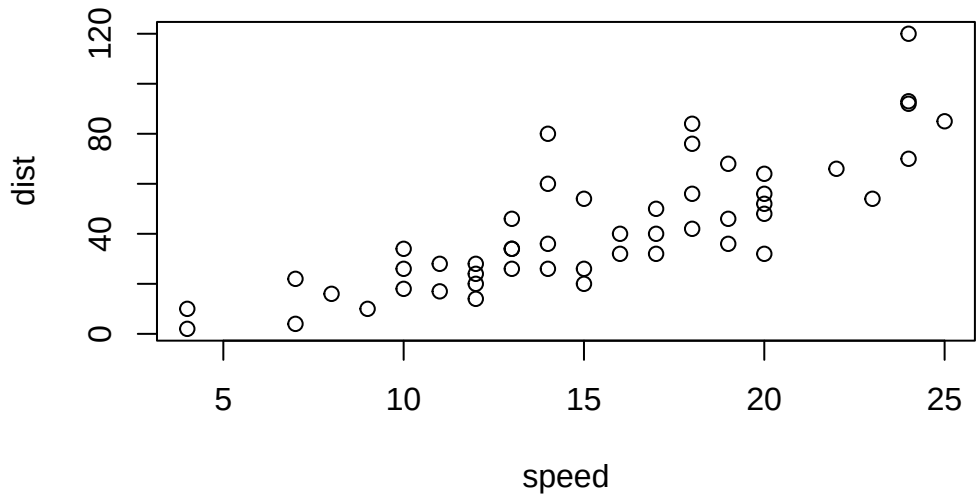
Comparing base graphics with ggplot

We can use the in-build data frame `cars` to play with plotting:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Now with ggplot: First I need to install the package with `install.packages()`

> I do not want to run `install.packages("ggplot2")` in my quarto document as this will result me me -re-installing the package every time I render my report.

Every time I want to use a function from a given package I need to load it up with a `library()` call:

```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

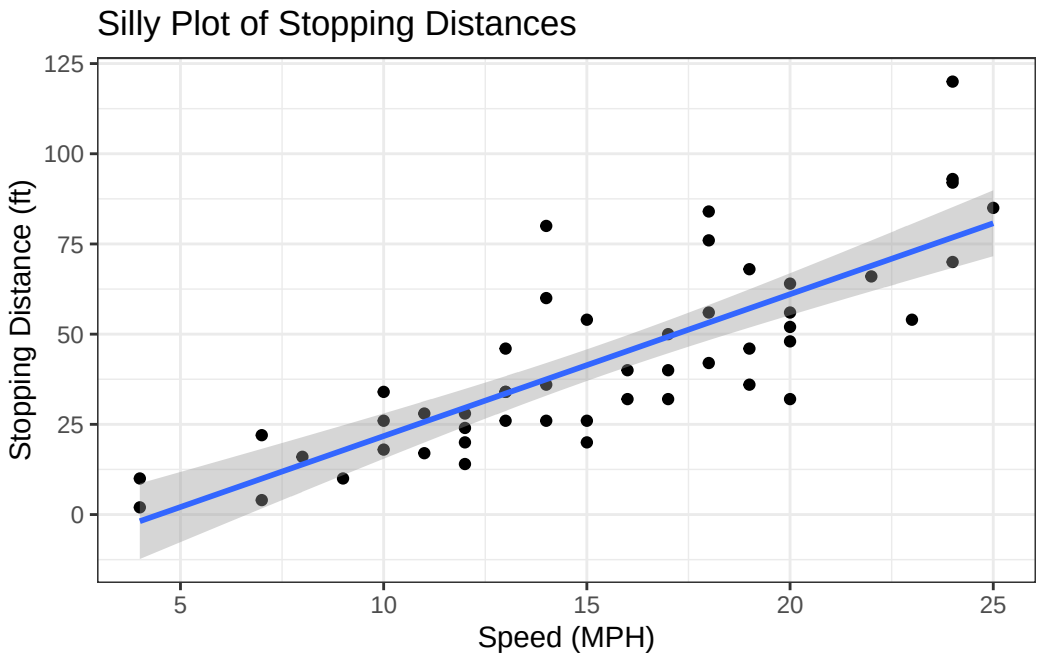
```
ggplot(cars)
```

Every ggplot needs at least 3 things:

- the **data** (data.frame we want to plot from)
- the **aesthetics** (how the columns of the data map to the plot)
- the **geom** (how you want the plot drawn, points, lines, etc.)

```
ggplot(data=cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm") +
  theme_bw() +
  labs(x="Speed (MPH)",
       y="Stopping Distance (ft)",
       title="Silly Plot of Stopping Distances")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



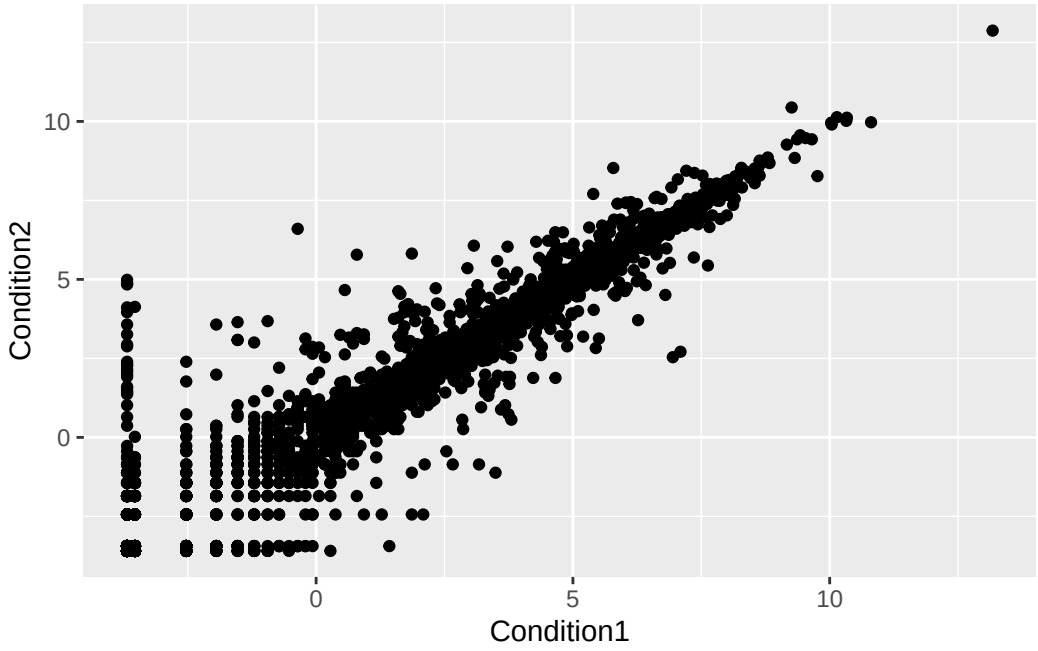
Gene Expression Plot

Read the data from online:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point()
```



Q. How many genes are in this dataset?

```
nrow(genes)
```

```
[1] 5196
```

Q. How many are “up” regulated?

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
#genes$State = "up"
#genes$State
```

```
sum( genes$State == "up" )
```

```
[1] 127
```

Q. What precentage of genes are “down” regulated?

```
n.total <- nrow(genes)
n.down  <- sum( genes$State == "down" )

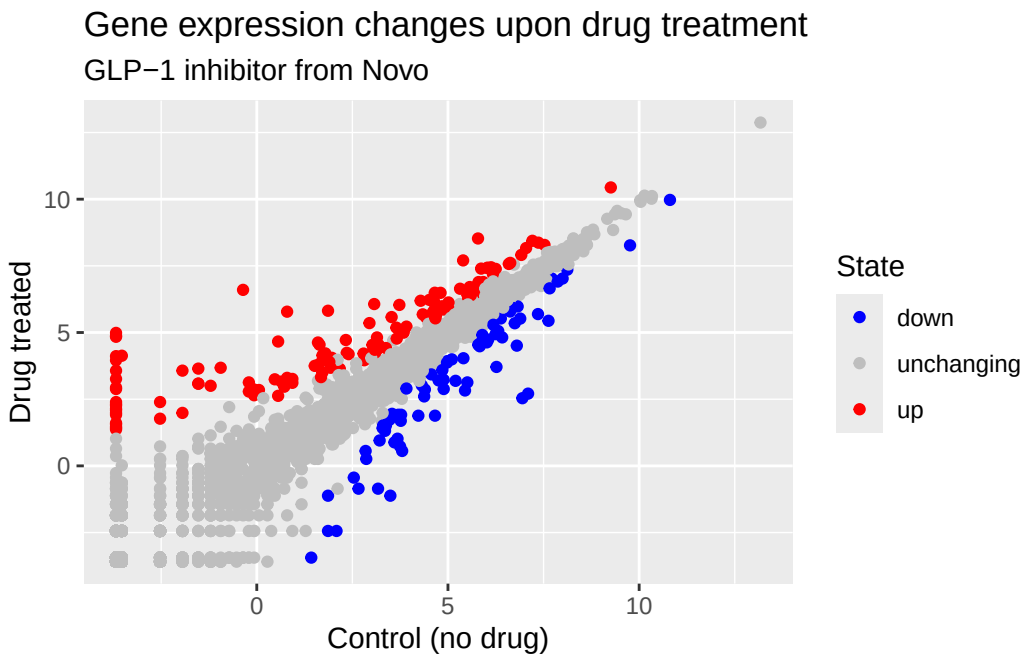
ans <- round( n.down/n.total * 100, 4)
ans
```

```
[1] 1.3857
```

The percentage of down genes is 1.3857 % and that is that….

```
myplot <- ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual( values=c("blue","gray","red") ) +
  labs(x="Control (no drug)",
       y="Drug treated",
       title="Gene expression changes upon drug treatment",
       subtitle = "GLP-1 inhibitor from Novo")
```

```
myplot
```



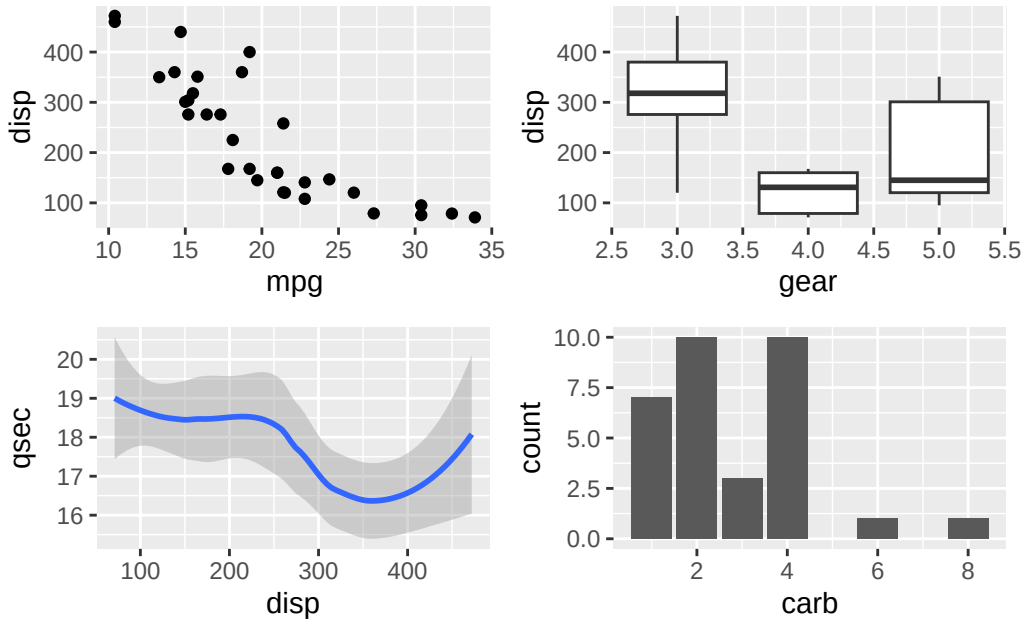
We can install extra packages with `install.packages()` like theam packages for spongebob colors or actually useful ones like **patchwork** for combining plots:

```
library(patchwork)

p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
p3 <- ggplot(mtcars) + geom_smooth(aes(disp, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))

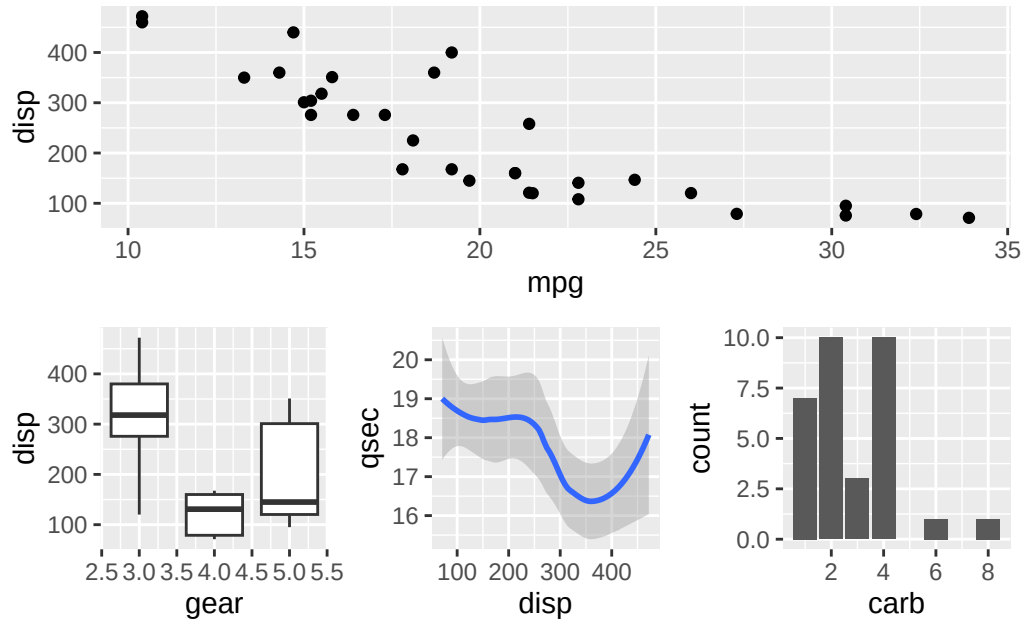
# Use patchwork to combine them here:
(p1 | p2) /
(p3 | p4)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

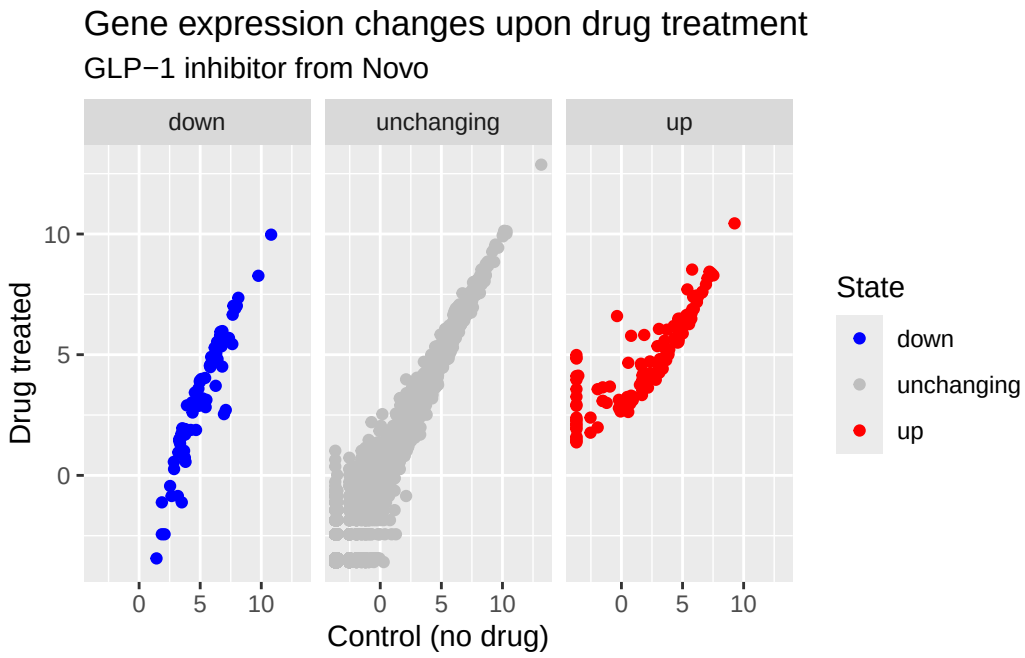



```
p1 / (p2 | p3 | p4)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
myplot +
  facet_wrap(~State)
```



The GapMinder data

The gapmider dataset is rather famous and has lots of stuff we can plot ;-)

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.ts
gapminder <- read.delim(url)
```

Have a wee peak:

```
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many data entries are there in this dataset?

```
nrow(gapminder)
```

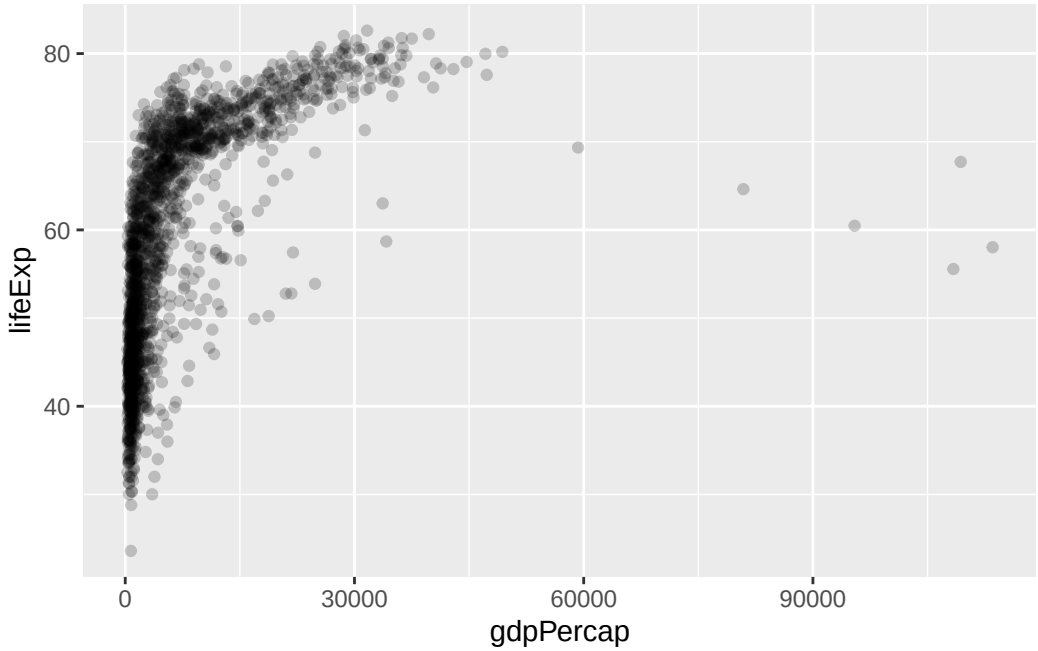
```
[1] 1704
```

Q. How many countries are in this dataset?

```
length( unique( gapminder$country ) )
```

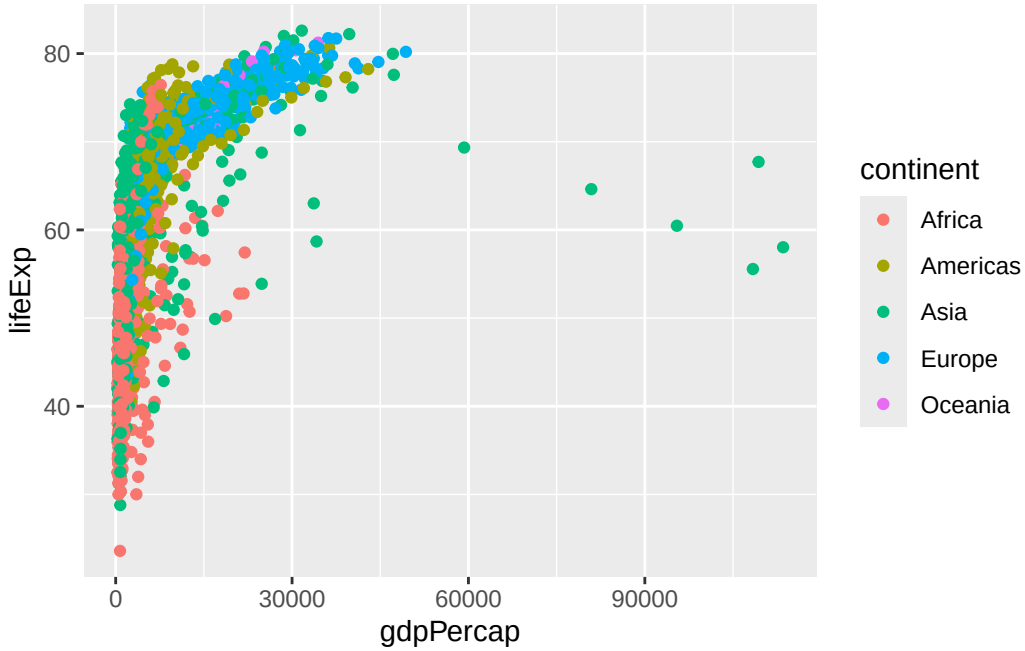
```
[1] 142
```

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp) +
  geom_point(alpha=0.2)
```



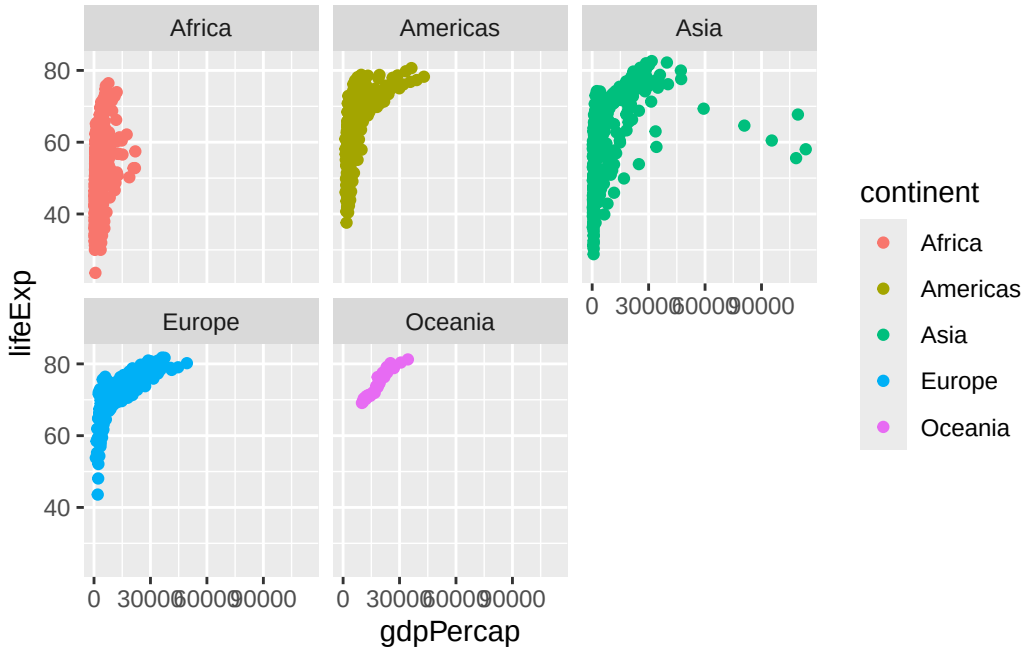
Add some color….

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
  geom_point()
```



Break out a facet per continent

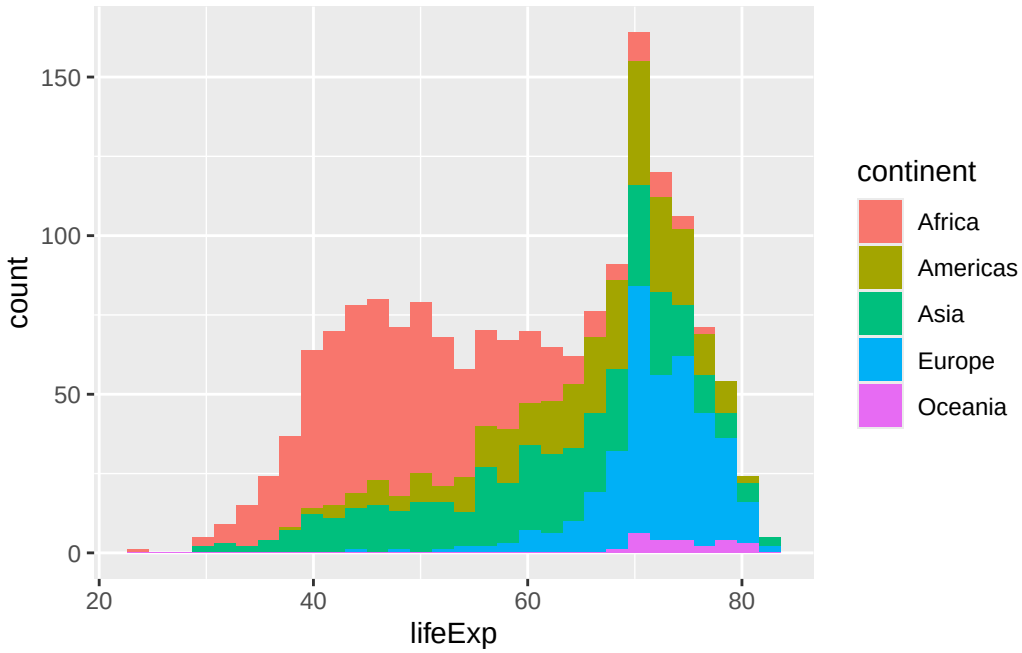
```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
  geom_point() +
  facet_wrap(~continent)
```



Histograms of

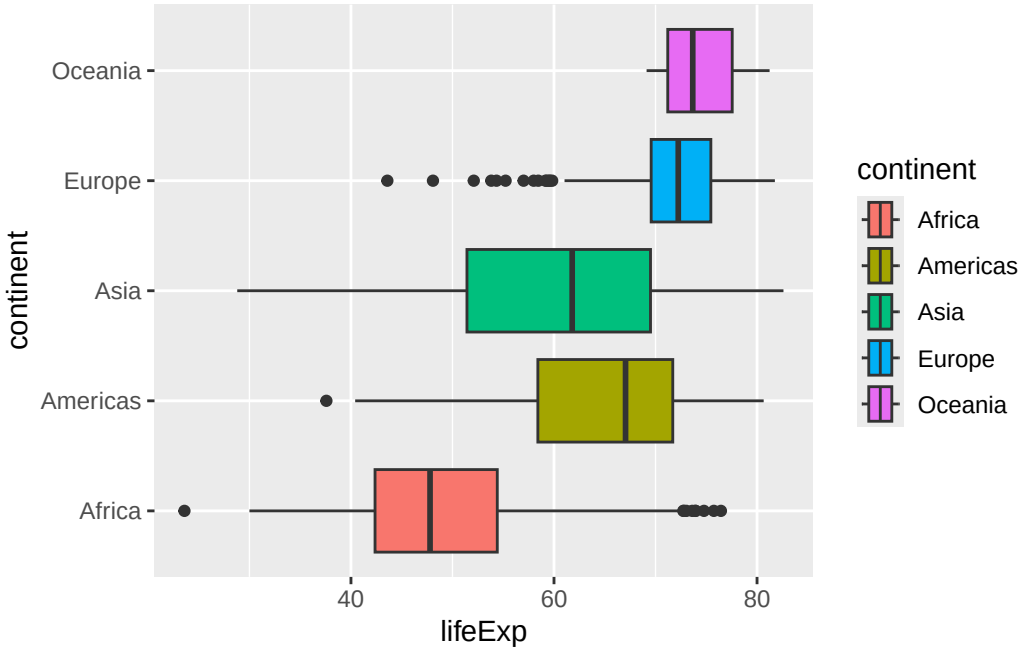
```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



and a boxplot

```
ggplot(gapminder) +
  aes(x=lifeExp, continent, fill=continent) +
  geom_boxplot()
```


Oceania
Europe
Asia
Americas
Africa
continent
40
60
80
lifeExp
continent
Africa
Americas
Asia
Europe
Oceania